



The use of choice experiments in the analysis of tourist preferences for ecotourism development in Costa Rica

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Many nations promote nature-based tourism in order to promote the dual goals of nature conservation and income generation. To be most effective in providing services that facilitate achievement of these goals, decision makers will need to understand and incorporate tourist preferences for nature appreciation, infrastructure, use restrictions, and other attributes of national parks and protected areas. This paper presents the use of choice experiments as a mechanism to analyze preferences of national and international tourists in relation to the development of Barva Volcano Area in Costa Rica. In this section of the Braulio Carrillo National Park, managers are faced with an immediate need to plan for greatly increased visitation rates due to a new road, which will greatly improve access. Choice sets were developed in collaboration with park managers. A survey was conducted of 171 Costa Rican and 271 foreign tourists who visited Poás Volcano, a well-visited alternative site to Barva Volcano. Survey data was analyzed using conditional multinomial logit models. Results of the study demonstrate, that both sets of tourists preferred: (i) improved infrastructure; (ii) aerial trams with observation towers and picnic areas; (iii) more information; and (iv) low entrance fees. Foreign tourists demonstrated strong preferences for the inclusion of restrictions in the access to some trails, whereas Costa Ricans did not show any significant preference for restrictions. Marginal willingness-to-pay for greater information was estimated to be \$1.54 for foreign tourists and \$1.01 for Costa Rican visitors. The study concludes that choice experiments are a useful tool in the analyses of tourist preferences for the development of protected areas in developing countries.

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Introduction

Many nations promote nature-based tourism in order to promote the dual goals of nature conservation and income generation. Many are expanding their systems of national parks and protected areas in order to protect biodiversity and other ecological services. It is hoped that tourist dollars can generate income for local populations and contribute to the

financial self-sufficiency of protected areas. Indeed, because of competing uses of lands among local populations and limited national government funds and donor assistance, the financial security of the local population and the financial self-sufficiency of these protected areas is crucial for their sustainability.

A key consideration in the strategy to attract tourists to protected areas is the possible reduction in biodiversity and nature conservation through litter, noise, and human access to fragile lands (Briassoulis, 1992). Thus, managers of protected areas need to balance the need to maintain

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ecological quality with the goal that these protected areas become financially self-sufficient. However, although many tourists prefer easy access and comfort, others accept additional travel time and expense in order to visit truly pristine zones in relative isolation. Thus, protected area managers must be able to develop infrastructure, access, and use restrictions, which facilitate the dual goals of conservation and tourism development. In order to be most effective in providing services that facilitate these goals decision makers and managers will need to understand and incorporate tourist preferences for nature appreciation, infrastructure, use restrictions, and other attributes of national parks and protected areas.

This analysis of tourist preferences will use the same random utility theory developed to estimate non-market values for environmental goods and services (Adamowicz *et al.*, 1998a). Specifically this study will present the use of choice experiments as a mechanism to analyze preferences of national and international tourists in relation to the development of Barva Volcano Area in Costa Rica. In this area of the Braulio Carrillo National Park, managers are faced with an immediate need to plan for greatly increased visitation rates due to improved access. Thus, park managers need to develop new infrastructure and use regulations in order to meet a possible twenty-fold increase in tourist visits.

The second part of this paper will present an overview of nature-based tourism in Costa Rica, which will be followed by background information on the Barva Volcano area of Braulio Carrillo National Park. The fourth section will present the use of choice experiments in the analyses of tourist preferences. The fifth section will present an analysis of national and foreign tourist preferences for different scenarios in the development of the Barva Volcano area. The paper will conclude with a discussion of these results and observations pertinent to protected area management and to the use of choice experiments.

Nature based tourism in Costa Rica

During the past decade Costa Rica has successfully promoted its tourist industry with an image of natural beauty and environmental consciousness, and has become one of the most popular ecological tourism destinations in the world (La Nación, 2000). Its promotion campaign stresses nature with the slogan 'Costa Rica: All Natural Ingredients'. Part of

the industry's success, which currently is the nation's most important source of foreign exchange, comes from its system of National Parks and protected areas.

Because of the success of Costa Rica's tourism industry, the dual goals of nature conservation and income generation are most often complementary. Nature conservation contributes to the tourism industry and tourism contributes to the national economy. Furthermore, tourism contributes to the National Park System of the Ministry of the Environment and Energy (MINAE), and the regional conservation areas through the entrance fees to the system of National Parks. Income from the national parks with high visitation rates is utilized for the protection and management of the entire national park system (Chase *et al.*, 1998).

Entrance fees to Costa Rican National Parks are somewhat controversial. As recently as 1994, Costa Rica had charged nominal entrance fees of US \$1.25 to both national and international visitors. Based on pressure from both the environmental community and fiscal authorities, entrance fees for foreign tourists were raised to \$15 (Chase *et al.*, 1998). This change in fees was not based on objective economic analysis (Schultz *et al.*, 1998). However, exceptions were made so that entrance fees purchased through tour operators were \$5 and fees purchased in advance were \$10. The result of this significant increase in entrance fees and the differentiated fee structure resulted in selective enforcement of entrance fees, a thriving black market in tickets, protests from local communities dependent on tourist visits, and a shift toward visitation of private nature reserves (Chase *et al.*, 1998). Eventually fees were reduced and standardized across national parks. During 1999, entrance fees for all national parks in Costa Rica were \$6.00 for foreign tourists and approximately \$1.25 for nationals, residents, and Central Americans. However, contingent valuation estimates of willingness to pay of Costa Rican and foreign tourists for park entrance, with some improvements to the parks, are considerably greater than the current fees (Schultz *et al.*, 1998).

Despite the fact that Costa Rica provides distinct nature-based tourist experiences throughout the country, tourist visits to National Parks are highly concentrated in a few sites (see Table 1). The three most popular tourist destinations, Manuel Antonio National Park (NP), Poás Volcano NP, and Irazu Volcano NP, receive about two thirds of the visitors who come to Costa Rica (Aylward *et al.*, 1996). An important determinant of visitation to national parks and protected areas is the travel time from San Jose, Costa Rica's capital, major population

Table 1. Visitation to selected national parks in Costa Rica

National Park	Visitors in thousands	Travel time hours*	Extension kilometers ²
Volcán poás	174.4	1.5	56
Braulio Carrillo	11.5	1.5	440
Volcán Irazú	111.8	1.0	23
Manuel Antonio	191.5	3.5	7

* Hours from San Jose in tourist bus.

center, and site of the principle international airport (Amelung, 1997).

Both Poás Volcano NP and Irazu Volcano NP are within two hours of travel time from San Jose in a tourist bus. This makes them convenient destinations for day trips organized by tourist companies for international tourists as well as trips for Costa Rican families and groups. Because of easy access, these two areas both have excessive visitation rates, i.e. the rates exceed their carrying capacity (Weaver, 1994).

The Barva Volcano sector of the Braulio Carrillo National Park

Due in part to the excess visitation at the Poás Volcano NP and Irazu Volcano NP among other factors, managers from the Central Volcanic Corridor Conservation Area (ACCVC) have decided to facilitate access to the Barva Volcano. Currently access to the site is by a steep muddy road three kilometers from the nearest pavement in the community of Sacramento. During the dry season months of January through March, four-wheel drive vehicles can reach the National Park entrance. During the rest of the year visitors are required to hike through often muddy terrain. Because of these restrictions, only an average of 912 visitors per month arrived during the peak dry season of January through March in 1998. This number was reduced to an average of 337 visits per month during the rest of 1998 (which is the only year where a complete set of visitation data was available). Because Barva Volcano offers many of the same attractions as Poás and Irazu and because travel time to the area will be similar to that of Poás once the final three kilometer stretch of road is paved, it is expected that visitorship at Barva could increase significantly to equal that of Poás which receives more than 20 000 visits per month during peak periods.

Braulio Carrillo NP is located in the center of Costa Rica, adjacent to the northern suburbs of

San Jose. At 440 000 square kilometers it is the nation's second largest national park. The park contains five distinct bio-climatic zones including tropical rain forest and mountain rain forests. The park extends from the peak of the Barva Volcano at 2906 m above sea level to 240 m above sea level. Rainfall averages over 3500 mm annually. Thus Braulio Carrillo NP contains the source of important rivers and much of the potable water supply for Costa Rica's densely populated Central Valley (Centro Científico Tropical, 1982). Most of the park has been managed for conservation in order to protect the watershed as well as the habitat for a variety of wildlife species including pumas, jaguars, tapirs, quetzals, and eagles. Given that one of the nation's principle highways traverses the park, it is extremely accessible. However, aside from a few rest centers and short trails, use of the park for tourism and recreation is restricted.

The Barva Volcano Sector is one of three small isolated areas within Braulio Carrillo NP designated for tourism, recreation, and educational visits. This area features seven kilometers of trails through tropical rain forest and secondary forest. These trails lead to two small lakes, Lake Barva at the volcano crater and Lake Copey. Because of acidic water, neither of these is suitable for swimming or fishing. Barva is home to an assortment of key species including the peccary, coyotes, and quetzals. Current infrastructure is very basic and rustic. Water is available, however it is not potable. There is a small unimproved picnic area at the park entrance, and managers allow camping at the site.

The entrance to the Barva Volcano sector is at the boundary of the National Park. Immediately adjacent to the park entrance is private land located within the Central Volcanic Corridor Forest Reserve. Because it is within the area designated as a buffer zone, the ACCVC maintains some control over land use. Thus ACCVC managers are hoping that basic tourist services, including parking and lodging, will be provided by the private sector once completion of the access road increases visitation facilities.

The use of choice experiments in the analyses of tourist preferences

Choice experiments were first employed in market and transport analysis. Recently it has been used to estimate the economic value of environmental goods

and services. As a valuation methodology, choice experiments can be considered to be a more generalized version of the single attribute dichotomous choice contingent valuation method. Indeed both choice experiments and contingent valuation can be classified as stated preference valuation methods (Adamowicz *et al.*, 1998a).

Although similar to conjoint analysis, which has been used extensively in marketing research, choice experiments have more direct links with economic theory. The method is based on Lancasterian consumer theory as well as random utility theory (Adamowicz *et al.*, 1998a, 1999). Lancaster's theory of derived utility proposes that consumers' utilities are defined over a bundle of attributes or characteristics of a purchased good or service (Gravelle and Rees, 1992). Thus a visit to a national park could imply the consumption of attributes such as fresh air, exercise, education, appreciation of nature, and observation of wildlife. A consumer will derive utility from these attributes as well as disutility from other attributes, such as congestion and physical exertion. Also, within a consumer choice framework, price can be considered to be another attribute. By soliciting consumer preferences for distinct hypothetical packages featuring different levels of each attribute, including price, welfare measures and values can be estimated.

As noted above choice experiments have been increasingly employed to analyze user preferences for environmental resources and to estimate the value of non-market goods and services, in developed countries. Adamowicz *et al.* (1998b) proposed alternative scenarios for the wildlife management in Alberta, and estimated the value of changes in the caribou population. The authors concluded that choice experiments had certain advantages over contingent valuation (CVM). In a similar study of moose hunting, Boxall *et al.* (1996) found that although there were significant differences in the values estimated by CVM and choice experiments, choice experiments might be more appropriate when substitution effects were important. Morrison *et al.* (1998) analyzed different scenarios for the management of the wetlands of the Macquarie River in New South Wales, Australia, and concluded that choice experiments produced acceptable estimates of non-use values. Hanley *et al.* (1998) reviewed the use of choice experiments and concluded that the methodology has advantages over contingent valuation in the valuation of characteristic values and in modeling substitution. Blamey *et al.* (2000) analyzed preferences for grazing restrictions and desert land management in Queensland, Australia and discovered certain differences between labeled

options and generic options. And Lindberg *et al.* (1999) used choice modeling to analyze preferences of Danish residents for the impacts of tourism development, and estimated the tradeoffs between negative impacts of increased traffic and rubbish and the positive impacts of job creation and tax reduction.

Economic analysis of tourist preferences in Costa Rica has focused on contingent valuation studies of park entrance fees. Schultz *et al.* (1997) tested the applicability of contingent valuation to estimate the willingness to pay for park entrance and concluded that the true willingness to pay for visitation to Poás and Manuel Antonio National Parks was considerably greater than the current \$6.00 for foreign visitors. Chase *et al.* (1998) also estimated a willingness to pay substantially greater than the current fee, and simulated a revenue maximizing system of differentiated fees. Echeverría *et al.* (1995) estimated the annual willingness to pay for preserving the Monteverde Cloud Forest Reserve and concluded that Costa Ricans valued the preserve more than foreign visitors. In a related study Lindberg and Aylward (1999) analyzed entrance into Poás, Irazu, and Manuel Antonio National Parks before and after a US \$1.00 fee increase and demonstrated an inelastic price elasticity of demand.

An analysis of user preferences for the development of Barva Volcano

Methodology

As noted park managers at the ACCVC are faced with a decision of how best to develop a site that is currently servicing 6500 visitors per year to meet the needs of a possible 150 000 visitors annually. In order to determine the most important attributes for analysis, a series of consultations were held with park managers from the ACCVC, tour operators, local experts in protected area management, non-governmental organizations, and scientists. Although final selection of attributes was made by the research team, attribute selection was restricted to issues of immediate importance to decision makers at the ACCVC. For instance, given the political sensitivity to changes in entrance fees, the price attribute was restricted to a 33% increase or decrease from current levels. Increased access to more remote areas of the Park were not considered,

nor were camping and lodging options, which are expected to be located outside of the Park entrance. Efforts were made to place a direct link between recreational and tourist use and conservation. For instance, although a recent study indicates that a large number of tourists, using current trails, could threaten the quetzal population (Universidad Nacional, 1992), there was not enough information on current quetzal populations to use this as an attribute. **Given these restrictions on the attributes to be analyzed, the emphasis of this research was not valuation of environmental goods and services, but on the determination of user preferences for the development of the Barva Volcano area.**

Once the preferred attributes were selected they were further reduced to accommodate design specifications. Table 2 presents the selected attributes. The development of infrastructure was considered to be an important attribute because of the imminent decisions of the ACCVC. Infrastructure also is

an important concern in the development of ecotourism, which aims to provide comfort with a low ecological impact (Anderson, 1993). Information is another attribute of particular concern to park managers and experts who prioritize environmental education. Rest areas and observation aids were combined in the experimental design phase of the investigation. ACCVC managers are considering the possibility of an aerial tram, shown to respondents in the information package as a ski lift, that traverses the forest at treetop level. This together with observation towers to facilitate appreciation of the rain forest from different heights are perceived to be an attractive option for tourists. Restricted access to certain trails, to reduce congestion and ensure the maintenance of a fragile ecosystem, are considered to be a mechanism to manage carrying capacity and allow certain tourists to enjoy less congested areas.

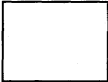
The original set of attributes allowed for ($3^3 \times 2^3$) possible combinations of attributes and levels. This

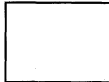
Table 2. Attributes and levels used in choice experiment

Variable short name	Description	Levels	
Infrastructure	The type of buildings and comforts at the visitors' center, and the development of trails	Rustic	Current status of Barva Volcano. Few improvements to trails to facilitate hiking. Construction with natural materials. Few comforts
		Semi-rustic	Current status of Irazu Volcano. Some improvement to trails to facilitate hiking. Construction with natural and modern materials. Some comforts
		Modern	Current status of Poás Volcano. Well groomed trails. Modern Construction. Comforts.
Information	Type of written and oral information presented	Minimum Medium Full	Interpretive signs on the trails Interpretive signs and pamphlets Signs, pamphlets, and information center with video presentations
View	Location of picnic areas and the availability of aerial trams and observation towers	No observation aids	No aerial trams nor observation towers and picnic at the Park entrance as well as along the trails
		Observation aids	Aerial trams and observation towers and picnic areas only at the Park entrance
Use Restrictions	Access restrictions	No restrictions Some restrictions	No restrictions to use of the trails Restricted access to some trails
Price	Entrance fee for Costa Rican and foreign visitors	Low	Less than current \$4.00 for foreign tourists and 350 Colons for Costa Ricans ^a
		Medium	Current \$6.00 for foreign tourists and 550 Colons for Costa Ricans
		High	Higher than current \$8.00 for foreign tourists and 750 Colons for Costa Ricans

^a The exchange rate in May of 1999 was 283.5 Colons for a US Dollar.

ATTRIBUTE	OPTION A	OPTION B	OPTION C
INFORMATION	Only interpretative signs on the trails	Video at an information center, pamphlets and signs	Video at an information center, pamphlets and signs
VIEW	With observation towers and tram	With observation towers and tram	Without observation tower nor tram
REST SITES AND PICNIC AREAS	Only at the park entrance	Only at the park entrance	At the park entrance and on the trails
INFRASTRUCTURE AND TRAILS	Rustic	Semi rustic	Modern
RESTRICTIONS OF USE	Limit to the number of people who enter certain trails	Without use restrictions	Without use restrictions
PRICE (Dollars)	4	6	8

OPTION A 

OPTION B 

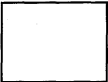
OPTION C 

Figure 1. A representative choice set.

was reduced manually using an experimental factorial design. The first step was the elimination of certain illogical combinations, such as low price and full information. This elimination was symmetric in order to preserve orthogonality. Later a series of orthogonal partitions were made reducing the number of attribute combinations to 36 (see Salinas, 1999 for details). The orthogonal elimination is critical because it is a sufficient condition to ensure that the assumption of independence of irrelevant alternatives is not violated (Louviere and Woodworth, 1983). This in turn is a direct reflection of independently and identically distributed error terms that are necessary in order to use multinomial logit estimation of the model (Adamowicz *et al.*, 1998a). An unintended consequence of this experimental design was the joining of the Rest Area and Observation Aids attributes. The 36 choice profiles were divided into three equal groups of 12. Because infrastructure was considered to be the most important of the attributes, the choice sites were blocked so that each set contained one choice with modern, one with semi-rustic, and one with rustic. This produced 1728 possible choice sets. In order to present each of the choice sets, without replication, there were a total of 432 questionnaires prepared and each respondent was asked to complete four choice sets.

Once the set of attributes and levels were defined an information package was developed, in Spanish and English, to present to surveyed tourists. This

package used photographs to assist the respondent in understanding both the current situation at the Barva Volcano site and the proposed attributes and levels. A survey instrument was prepared in both Spanish and English. The survey instrument included: (i) diagnostic and attitudinal questions to identify outliers as well as to introduce the respondent to the problem; (ii) questions to establish socioeconomic characteristics of the respondent; and (iii) a series of four choice sets to record tourist preferences among a set of three attribute groups. Figure 1 shows an example of a choice set.

The survey was conducted at Poás Volcano during May, June, and July of 1999. These months represented both low and peak seasons for tourist visits. Poás Volcano was selected for two reasons: (i) the extremely low visitation to Barva Volcano makes surveying inefficient; and (ii) visitors to Poás are the target group that are expected to visit Barva when it becomes an accessible alternative site. Furthermore, since the interviewed sample had all visited a volcano in central Costa Rica, a 'no visit' option was not included in the choice experiment, as suggested by Adamowicz (1998a).¹ Spanish and English speaking survey enumerators were recruited

¹ Three different anonymous reviewers questioned the omission of the 'no visit' option. Although the respondents had each visited a volcano, it is possible that some would choose not to visit any of the alternatives presented in the choice set. Further studies should be cautious in the omission of the 'no visit' option.

from among university students and under-employed tour guides. These enumerators were trained and supervised by the principle researcher. Interviews were conducted at the visitors' center at the Poás Volcano.

A key observation presented during the initial interviews with Park managers and experts was that the preferences of national and foreign tourists for park development were expected to be quite distinct. Costa Rican tourists were expected to prefer more facilities for family recreation while international tourists were expected to prefer an uncongested opportunity to enjoy natural beauty and biodiversity. For this reason it was important that both types of visitors be surveyed. As a result, 271 foreign and 161 Costa Rican tourists were randomly selected and surveyed. This distribution conformed to the percentage of foreign and national visitors who do visit Poás (Fernandez, 1999).

The collected data was analyzed in order to estimate the probability of choosing an alternative A from a choice set $\Phi = (A, B, C)$. Following from the previously presented Lancasterian theory and the random utility theory which is presented in numerous sources (Adamowicz, 1998a, 1999), the systematic component of the utility function is assumed to be linear and estimated as:

$$V_j = \alpha_1 \text{semi-rustic}_j + \alpha_2 \text{modern}_j + \beta_j \text{information}_j + \beta_2 \text{view} + \beta_3 \text{use}_j + \beta_4 \text{price}_j \quad (1)$$

with α and β representing the vector of coefficients of the explanatory levels and attributes from which utility is derived. Assuming independently and identically distributed Type 1 extreme value error terms with a scale factor $\mu > 0$ and $\sigma^2 = \pi^2/6\mu^2$, it is possible to use the multinomial logit model, such that the conditional probability is estimated as:

$$P(A|\Phi) = \frac{\exp(\mu V_A)}{\sum_j \exp(\mu V_j)} \quad \forall j \in \Phi. \quad (2)$$

This conditional logit model and marginal effects were estimated using LIMDEP, version 7.0 (see Greene, 1993, 1998).

Results and discussion

While the use of choice experiments in developed countries has recently been reported in a number of economics journals, published accounts of the use of choice experiments in developing countries have not

been encountered in the international literature. During the initial development of this investigation there was some concern among the experts interviewed that the methodology was too complex for certain interview respondents. Indeed, respondents without secondary education did demonstrate an inability to address a fairly complex hypothetical selection.² However, Costa Rica is a fairly well educated country with high levels of secondary and tertiary education (World Bank, 1996). Indeed 83% of the Costa Ricans and 99% of the foreign tourists interviewed had a secondary education. Furthermore, as recent contingent valuation studies of tourism in Costa Rica have shown, stated preference analysis is possible in Costa Rica.

Although the research was designed under the assumption that there are two distinct populations, national and foreign tourists, the survey instrument was identical for both groups, with the exception of the denomination of the price levels. Thus an initial test of the equality of the multinomial logit models was conducted under procedures outlined by Swait and Louviere (1993). The equality of the combined coefficients and scale parameters was rejected with a likelihood ratio test of the form:

$$\begin{aligned} & -2[\log\text{likelihood}(\text{pooled data}) \\ & \quad - \log\text{like}(\text{foreign}) - \log\text{like}(\text{national})] \\ & = 20.96 \sim \chi^2_7, \end{aligned} \quad (3)$$

where seven degrees of freedom for the six restricted parameters and the varying scale parameter μ . Given that the critical value for the χ^2 distribution at the 95% confidence level is 14.07 the equality of the combined coefficients and scale parameters between the two populations was rejected. The relative scale parameter $\mu_{\text{foreign}}/\mu_{\text{national}}$ was estimated to be 0.65, and the variables for the data from the survey of national tourists were adjusted. This enabled the comparison of the β coefficients of the two groups. The resultant likelihood ratio of 14.55 was calculated and compared to the critical value of 14.07. This test demonstrates that the two populations, Costa Rican and foreign tourists, represent different preference orderings. Thus, two models of preference orderings are presented.

Maximum likelihood estimates of the adjusted multinomial logit models are presented in Table 3.

² Surveys from respondents who were determined to be unable to understand the choice experiment were excluded from the final database and alternative respondents were surveyed. Many randomly selected tourists declined to participate in the survey, mainly because of time restrictions at the Poás site. No record of the initial response rate was reported.

Table 3. Results from multinomial logit model

Variable	Coefficient	Standard error	β /St error	$P[Z > z]$
Foreign tourists $n=271$				
Test for significance of model $\chi^2_4=32.527$			Significance for $\chi^2=1.00$	
Semi-rustic	0.7662	0.0905	8.464	0.0000
Modern	0.9781	0.0863	11.327	0.0000
Information	0.1350	0.0544	2.484	0.0130
View	0.1843	0.8246	2.235	0.0254
Use	0.2889	0.0780	3.703	0.0002
Price	-0.0874	0.0281	-3.104	0.0019
National tourists $n=161$				
Test for significance of model $\chi^2_4=31.73$			Significance for $\chi^2=1.00$	
Semi-rustic	0.3894	0.1126	3.459	0.0005
Modern	0.5306	0.1019	5.209	0.0000
Information	0.2434	0.0781	3.117	0.0018
View	0.4831	0.1042	4.634	0.0000
Use	0.1772	0.0972	1.823	0.0683
Price	-0.0008	0.0003	-2.558	0.0105

Table 4. Direct and cross elasticities of preferences

Choice Elasticity Attribute	Modern		Semi-rustic		Rustic	
	Direct	Cross	Direct	Cross	Direct	Cross
Foreign tourists $n=271$						
Information	0.143	-0.120	0.152	-0.093	0.222	-0.047
View	0.053	-0.048	0.079	-0.050	0.076	-0.017
Use	0.072	-0.069	0.080	-0.057	0.118	-0.028
Price	-0.195	0.156	-0.219	0.126	-0.290	0.059
Costa Rican tourists $n=161$						
Information	0.283	-0.200	0.370	-0.193	0.359	-0.129
View	0.101	-0.091	0.036	-0.030	0.165	-0.070
Use	0.049	-0.037	0.063	-0.034	0.062	-0.023
Price	-0.205	0.138	-0.235	0.118	-0.254	0.086

Each coefficient is of the expected sign, and all reach the 0.10 level of significance. The majority of these coefficients are highly significant. Semi-rustic and modern are coded as discrete changes in the infrastructure attribute and can be interpreted as dummy variables. Both view and use are two-level attributes so that the coefficients can be interpreted to reflect the discrete change from one level to the next (see Table 2). Information and price are three-level attributes and thus are coded to be continuous variables with the coefficients representing preferences for one increment in the level of the specified attribute.

The coefficient for price is negative in both populations reflecting a significant preference for lower entrance fees. Both semi-rustic and modern are highly significant in both models, demonstrating a preference for improved infrastructure among

both groups of visitors. Both view and information were positive and significant in both models. This implies that both groups of tourists had significant preferences for greater information, both on the trails and in the visitors' center. Also, both groups preferred greater visibility, with observation towers and an aerial tram with picnic areas at the park entrance. The coefficient for the USE attribute is positive and highly significant for the foreign tourists, but is not significant at the 95% probability level for the Costa Rican visitors. Thus whereas foreign tourists have a favorable view of limited access to certain trails, this preference is not highly significant among Costa Rican visitors.

As Adamowicz, Louviere and Swait (1998) argue, the coefficients from different models should not be directly compared. In order to facilitate the presentation and the comparison of these different models,

elasticities are calculated for the direct and cross effects of the probability of selecting an alternative with the change in an attribute level. Elasticities are presented in Table 4. They present the relative marginal change in probability of choosing an alternative (modern, semi-rustic, and rustic) with the relative marginal change in an attribute. Direct effects are calculated as:

$$\varepsilon_{X_{ik}}^{P(i)} = \frac{\partial P(i)}{\partial X_{ik}} \frac{X_{ik}}{P(i)} = \frac{\partial \ln P(i)}{\partial \ln X_{ik}} = [1 - P(i)] X_{ik} \beta_k. \quad (4)$$

Cross effects are calculated as:

$$\varepsilon_{X_{jk}}^{P(i)} = \frac{\partial P(i)}{\partial X_{jk}} \frac{X_{jk}}{P(i)} \quad \forall i \neq j. \quad (5)$$

In general, the calculated elasticities demonstrate that the marginal effects of the attributes on the selection of infrastructure are balanced. None of these attributes overwhelms the others. However, information has a relatively strong impact on the choices made by Costa Rican visitors. It is also shown that use restrictions have a minor impact on Costa Rican preferences, however it does have a greater impact upon the choice of foreign tourists. The strong influence of the price attribute for both Costa Rican and foreign visitors is shown.

Marginal values can be calculated from the marginal rate of substitution between a coefficient β_i and the coefficient for the price parameter, γ , so that marginal willingness to pay for an increase in an attribute is calculated as:

$$MWTP_i = -\frac{\beta_i}{\gamma}. \quad (6)$$

With the absolute value of γ representing the marginal utility of income (Alpizar *et al.*, 2001; Morrison *et al.*, 1998). Marginal willingness to pay (MWTP) for attributes and choices are presented in Table 5. Costa Rican willingness to pay (WTP) is adjusted to US dollars at the May 1999 exchange

rate of 283.5 colons per dollar. Costa Rican MWTP is generally lower than for foreign tourists because their actual and proposed entrance fees of 350, 550, and 750 colons (\$1.23, \$1.94, and \$2.65) is considerably lower than the \$4.00, \$6.00, and \$8.00 entrance fees proposed for foreign tourists. However, Costa Ricans present a relatively high MWTP for information. This reflects the fact that Costa Rican visitors often are with family and educational groups. Also, Costa Rican visitors often spend a few hours at the volcano with time available to appreciate the available information. In addition information, which is presented primarily in Spanish and secondarily in English, is often developed primarily for local clientele. In contrast, foreign visitors often arrive in tour groups, with limited time at the site and with guides who provide their own information.

Conclusions and observations

This study demonstrates that choice experiments are a feasible mechanism to analyze user preferences for the management of protected areas in developing countries. The study was facilitated by the fact that there is an immediate need for information to help park managers properly develop the Barva Volcano site. This reduced the hypothetical nature of the choice experiment and the correspondent biases. Furthermore, the implementation of the study was aided by the fact that a close substitute site, Poás Volcano, was a convenient location to conduct a survey, and that visitors to Poás can be considered to be informed consumers of the amenities in question.

The results of the study reveal that the preferences for national and foreign tourists are similar in direction and mostly similar in magnitude. Visitors preferred improved information as well as aerial trams and observation towers. Both modern and semi-rustic infrastructure were preferred to rustic infrastructure. In general, the survey respondents demonstrated a preference for site development, with efforts to provide more information, better views, and more modern infrastructure.

The preference for site development is in contrast to the fact that 72% of foreign tourists expressed concern that congestion did reduce the quality of their visit, but it is consistent with the indifference that Costa Rican visitors expressed to congestion effects. It may be difficult for National Park managers to develop use restrictions that address the concerns of foreign tourists, who want to appreciate

Table 5. Marginal willingness to pay for attributes in dollars

Attribute	Foreign tourists $n=271$		Costa Rican tourists $n=161$	
	Marginal WTP	Standard error	Marginal WTP	Standard error
Information	1.54*	0.57	1.01**	0.35
View	2.11	1.19	2.00*	0.84
Use	3.31*	1.40	0.73	0.49

**Significant at the 99% confidence interval.

*Significant at the 95% confidence interval.

nature in relative isolation, and Costa Rican visitors who would like a natural setting for family recreation and picnics. However, it should be noted that Costa Rican families tend to visit during weekends, whereas foreign tourists can schedule their visits during the week and encounter less congestion. If necessary it may be acceptable for Park managers to limit access to certain trails during the week, to allow for some isolated nature appreciation, and to allow unlimited access during weekends when crowds require more space.

Three concerns limited the use of this analysis in the estimation of economic values of environmental goods and services. The first was that the immediate needs of decision makers, which were a driving motivation of this study, did not include valuation. The second was the lack of appropriate biophysical information that would lead to a trade-off between appreciation of an environmental resource and price. The third was the limited range of the price attribute, due to the previously mentioned political constraints on modifying entrance fees. Neither of these constraints was particularly insurmountable at the time of the selection of attributes. However, the study was sufficiently motivated without a specified objective of valuation. In the future it may be necessary to consider an unfeasible range of prices in order to derive better estimates of welfare measures for valuation. However, an unfeasible range of prices does add to the possibility of misspecification bias on the part of the survey respondent (Mitchell and Carson, 1989).

Throughout tropical America, countries are attempting to utilize their forests, parks, beaches, and natural beauty to attract tourist dollars. Costa Rica has been very successful in marketing its nature-based tourist experiences and in providing tourist packages that meet international demand. As demand for tourism increases, both Costa Rica and its neighbors need to develop their natural areas to meet tourist demand and maintain environmental quality. Studies such as this may facilitate this development by informing decision makers of tourist preferences in order to best realize these combined goals.

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